

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) Semester IV Examination, 2025-26 (NEP)

BOTANY**Paper No. BOTMJ47 : Microbial Resources**

Ref. No.: Conf/Even/N/2025-26/22

Time: 2½ hours

Full Marks: 70

*(Write your Roll No. at the top immediately on the receipt of this question paper)**The figures in the right-hand margin indicate marks.**Answer **five** questions including Question 1 which is compulsory.***Question 1.** Answer the following in less than 100 words each: [7 × 2 = 14]

- (a) What do you understand by probiotics? Explain with a suitable example. [2]
- (b) What is the role of mashing in the production of alcoholic beverages? [2]
- (c) Define the term caseiculture. [2]
- (d) What is chelation? How does it help in phosphate solubilization? [2]
- (e) What is gene pyramiding? [2]
- (f) What is the role of ATP during nitrogen fixation? [2]
- (g) What is a “superbug” in bioremediation? [2]

Question 2. What are antibiotics? Describe their common modes of action with suitable example. How do microorganisms develop resistance to antibiotics? [14]**Question 3.** What do you understand by biotransformation? Explain the process with suitable examples. [14]**Question 4.** Explain the role of microorganisms in waste management with special reference to composting of solid wastes, wastewater treatment and methane production pathways. [14]**Question 5.** Discuss the process of biological nitrogen fixation with special reference to the structure and mechanism of the nitrogenase enzyme complex. [14]

Question 6. Answer **any two** of the following: [2 × 7 = 14]

- (a) Mechanism of action of *Bacillus thuringiensis* as a biopesticide. [7]
 - (b) Differentiate between *in situ* and *ex situ* bioremediation. [7]
 - (c) Microbial vaccines. [7]
-

Question 7. Answer **any two** of the following: [2 × 7 = 14]

- (a) Mechanisms of phosphate solubilization by Phosphate Solubilizing Bacteria (PSB). [7]
- (b) Explain the role of microorganisms in the production of organic acids with suitable examples. [7]
- (c) What is mushroom cultivation? Explain in detail the control measures required during mushroom cultivation. [7]